HACKATHON

Talent Match Hackathon

Build a small, real, end-to-end tool that helps a candidate understand how ready they actually are for a specific job.

24 OCT
Hackathon launch
9:30 AM sharp

29 OCT
Peer presentations
Top 9 selected

30 OCT
Expert panel
Top 3 selected

31 OCT
TGL Blitz
Grand finale

RADIX

PARTICIPANT BRIEF

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Everything your team needs to build, integrate and present **RADIX Talent Match**.

Read all of it — not just the part about your own role.

24 Oct	Launch · 9:30 AM
29 Oct	Peer presentations
30 Oct	Expert panel
31 Oct	TGL Blitz

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01 THE CHALLENGE

What you're building

RADIX Talent Match — a small, real, end-to-end tool that helps a candidate understand how ready they actually are for a specific job, built on the same 12-skillset framework and real hiring data RADIX already researched.

WHEN YOU'RE DONE, SOMEONE SHOULD BE ABLE TO SIT DOWN AT YOUR TOOL AND…

01
JD Analytics
Upload a JD and see what skills it actually requires.

02
Resume Parsing
Upload a resume and see what skills it shows.

03
Profile Builder
Build a full profile that goes beyond the resume — hackathons, certifications, preferred roles.

04
Talent Check
See how they stack up against a specific company's bar, across all 12 skillsets.

05
Skill Matching
See exactly how well they match one specific job posting, with a score and a gap list.

> Each of the five team members builds one of these five pieces. Then you integrate them into one working tool.

Why this matters

Everything RADIX has built so far has been the **company's side** of a placement platform. This hackathon flips the lens: same underlying skill-set data, same quality mindset — but built for the **candidate** trying to figure out if they're actually ready.

LEVELS 1–2
A 163-parameter research framework, a 12-skillset competency dictionary, and real hiring-round data for real companies.

LEVEL 3
Validating data automatically.

LEVELS 4–5
Generating and orchestrating data with agents.

THIS HACKATHON
The same data, built for the candidate.

02 HACKATHON TIMELINE

From launch to the TGL Blitz stage

24 OCT · SAT

9:30 AM SHARP

Hackathon launch

The hackathon is handed to students at 9:30 AM sharp. Form your team of five, assign the five roles, appoint your Scrum Master — and get started.

24–28 OCT

BUILD PHASE

Build, integrate, refine

Work closely together as a team and keep making real progress. There is ample time — use it to build every piece well and wire them into one working tool.

28 OCT · WED

DEADLINE

Everything complete

Your integrated tool and your presentation must be fully complete by 28 October.

29 OCT · THU

ROUND 1 · FIRST HALF

Peer presentations 45 minutes per team

Every team presents its hackathon work. This round is highly important — it decides which teams move forward.

→ Top 9 teams selected

30 OCT · FRI

ROUND 2 · FIRST HALF

Expert panel presentations 45 minutes per team

The top 9 teams present to a panel of:

Industry experts · TGL experts · JSS faculty · JSS management

→ Top 3 teams selected

31 OCT · SAT

GRAND FINALE

TGL Blitz

The top 3 teams showcase their hackathon work in the auditorium (details will be shared by the college), before a panel of:

JSS Noida management · Industry experts · Investor community & startups · TGL team

03 SELECTION & CERTIFICATES

Every participant is recognised

Every member of every team receives a certificate. Teams that progress through the presentation rounds earn higher recognition.



ALL TEAMS

Participation Certificate

For every single member who works on this hackathon, across all teams.

All participants



TOP 9 TEAMS

Pre-Finalist Certificate

For the 9 teams selected from the peer presentations, who go on to present to the expert panel.

Selected 29 October



TOP 3 TEAMS

Finalist Certificate

For the top 3 teams selected by the expert panel, who present at the TGL Blitz grand finale on 31 October.

Selected 30 October

YOUR PRESENTATION DECIDES YOUR PATH

45 minutes

per team — in the peer presentations on 29 October, and before the expert panel on 30 October if selected.

Plan a **live walkthrough of your integrated tool** — not just a slide deck — and rehearse it end to end. Cover:

01 **The problem**

02 **Your architecture**

03 **Live demo**

04 **One challenge you faced**

04 YOUR TEAM

Five members. Five roles. One Scrum Master.

Every team has exactly five members. Each member owns one of the five roles, and one member is also appointed Scrum Master.

+ SCRUM MASTER				
1	2	3	4	5
ROLE 1	ROLE 2	ROLE 3	ROLE 4	ROLE 5
JD Analytics	**Resume Parsing**	**Profile Builder**	**Talent Check**	**Skill Matching**

Illustrative — the Scrum Master can be any one of the five members. The team decides.

Exactly five

Every team has five members — no more, no fewer.

One role each

Each member owns and builds one of the five pieces.

One Scrum Master

Chosen by the team. Still owns and builds one role.

Read all five roles

Know what you hand off, and what is handed to you.

SCRUM MASTER Keeps the team together and on track

In addition to building their own role, the Scrum Master:

01 **Leads the kick-off**
Makes sure the shared data contract is agreed and written down where everyone can see it.

02 **Keeps the team on schedule**
Makes sure the team is progressing well and everything is complete by 28 October.

03 **Owns the milestone tracker**
Keeps every milestone's status current: Not Started, In Progress, Blocked or Done.

04 **Clears blockers early**
Watches the dependencies — Roles 4 and 5 must not be left waiting on Profile Builder.

05 **Leads integration**
Coordinates wiring the five pieces into one flow, and keeps known bugs logged.

06 **Gets the team presentation-ready**
Makes sure the walkthrough is prepared and rehearsed end to end.

05 THE FIVE ROLES

One role, one person, one piece

Read all five, not just your own — you need to know what you're handing off to, and what's being handed to you.

01 JD Analytics

LLM EXTRACTION

Given a JD (PDF or Word), extract a structured list of the skills and technologies it requires.

INPUT
One of the 6 sample JDs, or any JD, as a PDF or DOCX file.

OUTPUT
A structured skill list in the shared format.

APPROACH A strong LangChain/LLM use case. Feed the JD text to an LLM with a prompt that maps requirements onto the 12 RADIX skill categories plus any named technologies. Plain keyword-matching alone will miss a lot — the JDs are written in natural prose on purpose.

> The skill signal is concentrated in "Key Responsibilities" and "What We're Looking For" — start there.

02 Resume Parsing

LLM EXTRACTION

Given a resume (PDF or Word), extract key technologies, skills, and technical signal.

INPUT
One of the 4 sample resumes, or any resume, as a PDF or DOCX file.

OUTPUT
A structured skill list in the shared format, plus fields you can reliably pull (education, projects, experience).

APPROACH The same LLM-extraction pattern as JD Analytics, applied to a differently-shaped document. Talk to whoever's building #1 — you may be able to share real code, not just a data format.

> Resumes are messier and less standardized than JDs. Build in tolerance for format variation rather than assuming one layout.

03 Profile Builder

PLAIN CODE

Let a candidate build a full profile: basic details, skills, hackathons/internships, certifications, preferred roles, and a CV upload.

INPUT
Manual form entry, ideally pre-filled from Resume Parsing's output.

OUTPUT
A saved candidate profile in the shared format.

APPROACH Mostly good old coding — a form, basic validation, and a save/load mechanism (JSON). LangChain isn't really needed here, but you can use it creatively if you want to.

> Everyone else depends on this. Get a save/load loop working early, even with dummy data, so Roles 4 and 5 aren't blocked waiting on you.

04 Talent Check

PLAIN CODE

One button that compares a candidate's profile against a company's expected bar across all 12 RADIX skillsets, and shows the gap.

INPUT
A saved profile (from Profile Builder) and a selected company.

OUTPUT
A readiness score and a per-skillset gap breakdown.

APPROACH Mostly comparison and scoring logic — plain code. The company's 12-skillset expectations already exist in RADIX's real data; you'll be given a local snapshot of it for the 3 companies in scope.

> Decide early what "ready" means numerically — a threshold, a weighted score, whatever you choose. That's a design decision, not just an implementation detail, and it changes what the button actually shows.

05 Skill Matching

FUZZY MATCH

Show how well a candidate matches one specific JD — not a company in general, one actual posting.

INPUT
A profile or resume, plus one extracted JD skill list from Role 1.

OUTPUT
A match score, plus which skills matched and which are missing.

APPROACH Matching one skill list against another has real fuzziness — the same skill gets described differently in different documents. A good place to use an LLM or simple similarity matching if you have time; exact/fuzzy string matching if you don't.

> A match score with no explanation is much less useful than one that also says what's missing. Prioritize the gap list over the number.

06 THE SHARED DATA CONTRACT

One language for all five pieces

Below is a proposed shape for the data passing between your five pieces — a starting point, not a final spec. Your first job as a team is to review it, debate it, and lock in a version everyone can build against. Change field names if you want — just don't let the debate run long.

The 12 RADIX skill categories

Every skill is tagged with exactly one `category_code`.

COD	Coding	DSA	Data Structures & Algorithms	OOD	Object-Oriented Design
APTI	Aptitude	COMM	Communication	AI	AI
CLOUD	Cloud	SQL	SQL	SWE	Software Engineering
SYSD	System Design	NETW	Networking	OS	Operating Systems

Anything outside these twelve uses `OTHER`.

The building block: a single skill

<skill>

```
{
  "skill_name":     "...",
  "category_code": "DSA|COD|OOD|APTI|COMM|AI|CLOUD|SQL|SWE|SYSD|NETW|OS|OTHER",
  "evidence":       "short quote or reason",
  "confidence":     "high|medium|low"
}
```

skill_name
The skill or technology.

category_code
One of the 12 codes, or OTHER.

evidence
A short quote or reason.

confidence
High, medium or low.

Who hands what to whom

ROLE	TAKES IN	HANDS OFF
JD Analytics	A JD (PDF or DOCX)	JD skill list → Skill Matching
Resume Parsing	A resume (PDF or DOCX)	Resume skill list → Profile Builder
Profile Builder	Form entry, pre-filled from Resume Parsing	Saved profile → Talent Check, Skill Matching
Talent Check	Saved profile + selected company	Readiness score + per-skillset gaps
Skill Matching	Profile or resume + one JD skill list	Match score + matched / missing skills

What each piece hands off

Extracted skill list ROLE 1 ROLE 2

```
{
  "source_type": "jd|resume",
  "source_file": "...",
  "company":     "...",
  "role":        "...",
  "skills":      [ <skill>, ... ]
}
```

Candidate profile ROLE 3

```
{
  "name":            "...",
  "email":           "...",
  "education":       "...",
  "skills":          [ <skill>, ... ],
  "hackathons":      [...],
  "internships":     [...],
  "certifications":  [...],
  "preferred_roles": [...],
  "cv_file":         "..."
}
```

Talent Check result ROLE 4

```
{
  "company": "...",
  "skillset_gap": [ {
    "category_code":   "...",
    "required_level":  1-10,
    "candidate_level": 1-10,
    "gap":             true|false
  } ],
  "readiness_score": 0-100
}
```

Skill Matching result ROLE 5

```
{
  "jd_source_file": "...",
  "match_score":    0-100,
  "matched_skills": [...],
  "missing_skills": [...]
}
```

LOCK IT BEFORE YOU BUILD

Agree on the shared JSON schema for skills, profile and match, and write it down somewhere everyone can see.

07 YOUR TEST DATA

Build against real material

Your hackathon pack contains 6 real job descriptions across 3 companies and 4 sample resumes — each supplied in both PDF and Word (.docx) format.

JOB DESCRIPTIONS · 3 COMPANIES, 2 ROLES EACH

COMPANY	ROLE	LOCATION
Google LLC	Software Engineer (SWE) Data Scientist	Bengaluru, India (Hybrid) Bengaluru, India
Microsoft Corporation	Software Engineer Data Analyst	Hyderabad, India Hyderabad, India
Oracle Financial Services Software	Associate Software Engineer Application Support Analyst	Bengaluru / Mumbai, India Bengaluru / Chennai, India

SAMPLE CANDIDATES · DELIBERATELY DIFFERENT SKILL PROFILES

CANDIDATE	PROFILE	FIT, AS STATED IN THE RESUME
Ananya Rao	Systems-leaning generalist	Strong fit for SWE-heavy roles, weaker fit for data roles
Karthik Subramaniam	Data-leaning analyst	Strong fit for Data Scientist / Data Analyst roles, weaker fit for pure SWE roles
Priya Menon	Entry-level, support-oriented	Best fit for the Oracle FSS roles, weaker fit for Google / Microsoft
Rohan Verma	Strong all-rounder	Competitive fit across nearly every role in the pack

Because the profiles differ on purpose, **Talent Check and Skill Matching results should visibly differ from person to person.**

! TALENT CHECK NEEDS ONE MORE FILE

Each company's expected level across all 12 skillsets. You'll work from a local JSON snapshot of this data for the 3 companies rather than live Supabase — ask your facilitator for `talent_check_company_skillsets.json` if you don't see it in your starter folder.

08 HOW TO BUILD IT

From blank page to one working tool

1

Plan

WHOLE TEAM

Read this brief and the JD / resume pack. Lock the data contract, names and file locations. Everyone knows exactly what they're building and what shape their output takes.

2

Build

EACH MEMBER

Build your own piece. Test against the sample JDs and resumes as you go — don't wait until integration to discover your JSON doesn't parse.

3

Integrate

WHOLE TEAM

Bring all five pieces together into one working flow. It never goes perfectly on the first try — expect some “why won't this run together” time.

4

Prepare

WHOLE TEAM

Get the walkthrough ready and rehearse the live demo end to end, without a mentor's help.

THE INTEGRATED FLOW

Pick a JD › Parse a resume › Build / update a profile › Run Talent Check › Run Skill Matching

Track your progress

RADIX_Talent_Match_Milestone_Checklist.xlsx

Whole team	10 milestones
JD Analytics	2 milestones
Resume Parsing	2 milestones
Profile Builder	2 milestones
Talent Check	2 milestones
Skill Matching	2 milestones

Your milestone checklist breaks the work into clear milestones — each with its owner and a definition of done. The Scrum Master keeps it current; the Summary sheet shows progress by phase and by owner.

Not Started · In Progress · Blocked · Done

09 STACK & DEFINITION OF DONE

Build it right

Tech stack

LANGUAGE	**Python** throughout — same stack as the rest of RADIX.
AI LAYER	Use **LangChain** or an LLM agent wherever it genuinely helps — JD Analytics, Resume Parsing, and the fuzzy-matching part of Skill Matching are strong candidates. Where plain code is simpler and more reliable — Profile Builder's form, Talent Check's scoring logic — use plain code. An agent that's harder to debug than the problem it solves isn't a win.
FRONTEND	**React**, matching the RADIX portal's stack.
DATA STORAGE	**Supabase** — JSON for most things.

What "done" looks like

- **All 5 pieces run on their own** against at least 2 of the sample JDs / resumes.
- The 5 pieces are **wired into one flow**, end to end, for at least one company / JD / candidate combination.
- Talent Check and Skill Matching give **visibly different results** for different candidates — the scoring logic isn't a constant.

Your first steps on 24 October

- [ ] Team of five formed
- [ ] Scrum Master appointed
- [ ] Data contract agreed and written down
- [ ] All five roles assigned
- [ ] Everyone has read this brief and the JD / resume pack
- [ ] `talent_check_company_skillsets.json` in hand

RADIX TALENT MATCH HACKATHON

24 October 2026

9:30 AM sharp

Form your team of five. Appoint your Scrum Master. Get started.

RADIX